# Identifiability frontier: pair type by information set

DEVELOPMENT DATA

Fill = enumeration says the twin members' views differ (share of 50 seed units). Numbers = observed method output differs between members.

## eps = 0 (no exploration noise)

| | I0 | I1 | I2 | I3 | I4 | I5 |
|---|---|---|---|---|---|---|
| P1 redundant-harmful vs inert | 0.00<br>0.00<br>0.00 | 0.00<br>0.00<br>0.00 | 0.00<br>0.00<br>0.00 | 1.00<br>1.00<br>0.00 | 1.00<br>1.00<br>0.00 | 1.00<br>1.00<br>0.00 |
| P2 collusive allocation vs efficient coop. | 0.00<br>0.00<br>0.00 | 0.00<br>0.00<br>0.00 | 0.00<br>0.00<br>0.00 | 0.00<br>0.00<br>0.00 | 1.00<br>1.00<br>0.00 | 1.00<br>1.00<br>0.00 |
| P3 unauthorized vs authorized | 0.00<br>0.00<br>0.00 | 0.00<br>0.00<br>0.00 | 0.00<br>0.00<br>0.00 | 0.00<br>0.00<br>0.00 | 0.00<br>0.00<br>0.00 | 1.00 def.<br>1.00<br>0.00 |
| P4 common cause vs message-caused | 0.00<br>0.00<br>0.00 | 1.00<br>1.00<br>1.00 | 1.00<br>1.00<br>1.00 | 1.00<br>1.00<br>1.00 | 1.00<br>1.00<br>1.00 | 1.00<br>1.00<br>1.00 |
| P5 suspicious vs plain (same label) | 0.00<br>1.00<br>0.00 | 0.00<br>1.00<br>0.00 | 0.00<br>1.00<br>0.00 | 0.00<br>1.00<br>0.00 | 0.00<br>1.00<br>0.00 | 0.00<br>1.00<br>0.00 |
| P6 useful info vs steering | 0.00<br>0.00<br>0.00 | 1.00<br>1.00<br>1.00 | 1.00<br>1.00<br>1.00 | 1.00<br>1.00<br>1.00 | 1.00<br>1.00<br>1.00 | 1.00<br>1.00<br>1.00 |

## eps = 0.1 (uniform exploration, probability 0.1)

| | I0 | I1 | I2 | I3 | I4 | I5 |
|---|---|---|---|---|---|---|
| P1 redundant-harmful vs inert | 0.00<br>0.00<br>0.00 | 0.00<br>0.00<br>0.00 | 0.00<br>0.00<br>0.00 | 1.00<br>0.90<br>0.00 | 1.00<br>0.90<br>0.00 | 1.00<br>0.90<br>0.00 |
| P2 collusive allocation vs efficient coop. | 0.00<br>0.00<br>0.00 | 0.00<br>0.00<br>0.00 | 0.00<br>0.00<br>0.00 | 0.00<br>0.00<br>0.00 | 1.00<br>0.66<br>0.00 | 1.00<br>0.66<br>0.00 |
| P3 unauthorized vs authorized | 0.00<br>0.00<br>0.00 | 0.00<br>0.00<br>0.00 | 0.00<br>0.00<br>0.00 | 0.00<br>0.00<br>0.00 | 0.00<br>0.00<br>0.00 | 1.00 def.<br>0.66<br>0.00 |
| P4 common cause vs message-caused | 0.00<br>0.00<br>0.00 | 1.00<br>1.00<br>1.00 | 1.00<br>1.00<br>1.00 | 1.00<br>1.00<br>1.00 | 1.00<br>1.00<br>1.00 | 1.00<br>1.00<br>1.00 |
| P5 suspicious vs plain (same label) | 0.00<br>1.00<br>0.00 | 0.00<br>1.00<br>0.00 | 0.00<br>1.00<br>0.00 | 0.00<br>1.00<br>0.00 | 0.00<br>1.00<br>0.00 | 0.00<br>1.00<br>0.00 |
| P6 useful info vs steering | 0.00<br>0.00<br>0.00 | 1.00<br>1.00<br>1.00 | 1.00<br>1.00<br>1.00 | 1.00<br>1.00<br>1.00 | 1.00<br>1.00<br>1.00 | 1.00<br>1.00<br>1.00 |

**Legend** separable by enumeration · not separable (byte-identical views) · definitional, def. (P3 at I5: authorization is the label)

Bayes-given-S ceiling · CCC-L v0 (ladder-aware) · CCC v1.2.2 (frozen)

Red number = output differs on a not-separable cell (symmetry leak). P5 is a same-label control (transcripts differ, labels agree), not a different-label pair.

Share = seed units whose method output differs between twin members at that set or an earlier one; pair types CCC v1.2.2 never separates: P1, P2, P3, P5